

Intermediate Macro

Chapter 13: The Challenges of Growth

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Debatable Issues of Technological Progress

13.1 The Future of Technological Progress

- Chapter 10 through 12 established technological progress as the determinant of long-run growth per person.
- Nevertheless, workers often fear that technological progress will eliminate their jobs.
- In response to this concern, Chapter 13 argues that productivity growth does not appear systematically to increase the unemployment rate.
- However, technological progress may affect the distribution of income.
- Economic growth also comes at a cost to the planet.
- There are environmental trade-offs we must make to spur additional growth.
- The question of how to balance growth with environmental protections will continue to be a topic of debate among policy makers and economists.

A slowdown of Productivity Growth?

13.1 The Future of Technological Progress

- We might expect rapid technological progress to increase productivity.
- Yet, US measured productivity growth has slowed down since the mid-2000s, running at less than half the rate of earlier decades.
- One argument is that technological progress is hard to measure and that we also understate the rate of productivity growth.
- Some argue that the current major innovations are less important than the major innovations of the past.
- Major innovations are innovations that have applications in many fields and many products (For this reason, these innovations are called **general-purpose technologies.**)

General-Purpose Technologies

13.1 The Future of Technological Progress

- Robert Gordon, of Northwestern University, argues that the two major innovations of the last 150 years were electricity and the internal combustion engine.
- Others, in particular Eric Brynjolfsson of MIT, argue that digitization, together with the increasing power of computers, will transform our lives as much as or more than electricity or the internal combustion engine did earlier.
- Artificial intelligence and machine learning may transform nearly any activity, from driverless cars to the mapping of the human genome and major progress in medicine, to the way lawyers access relevant jurisprudence.
- He argues that the current low productivity growth reflects the slow process of diffusion and the discovery of new applications.

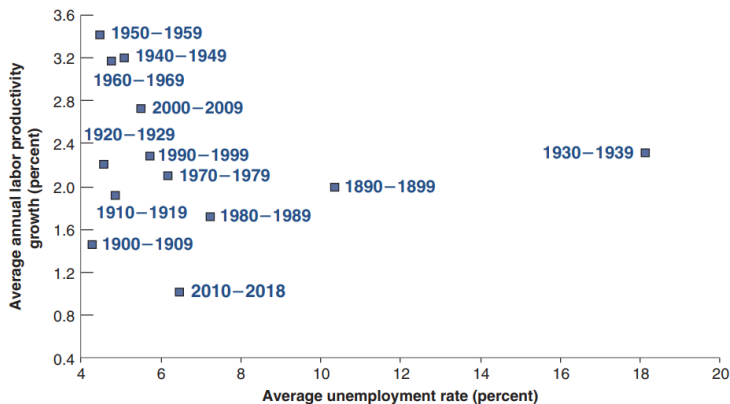
Technological Unemployment

13.2 Robots and Unemployment

- For the last 200 years, workers have worried they will be displaced by technology.
- We often see the theme of **technological unemployment** resurface when unemployment is high.
- From the machine-smashing *Luddites* in the early 19th century to the *technocracy movement* that appeared during the Great Depression, we often hear workers argue the high unemployment is due to the introduction of new machinery or technology.
- However, the data show this fear is unfounded.

Productivity Growth and Unemployment, Averages by Decade, 1980-2018

13.2 Robots and Unemployment



Caveat

13.2 Robots and Unemployment

- Periods of higher productivity growth are associated with the lowest unemployment rates.
- Technology is one of the primary drivers of increases in productivity. The only caveat is that we cannot be sure this relationship will hold in the future.
- Will robots replace workers and increase everyone's standard of living?
- or will they simply increase wealth inequality?
- Policy makers will need to watch closely to ensure everyone receives some minimum level of income in this future scenario.

The Increase in Wage Inequality

13.3 Growth, Churn, and Inequality

- Even if technological progress helps the economy generally and does not lead to increases in the aggregate unemployment rate, it still may hurt some individuals.
- In the United States, for example, wage inequality has increased substantially over the past 20 years.
- Workers with a high level of education have enjoyed an increase in their relative real wage; workers with the least amount of education have actually suffered a decrease in the absolute level of their real wage.
- The increase in the relative wage of high-skill workers reflects increased relative demand for high-skill workers.

13.3 Growth, Churn, and Inequality



The Causes of Increased Wage Inequality

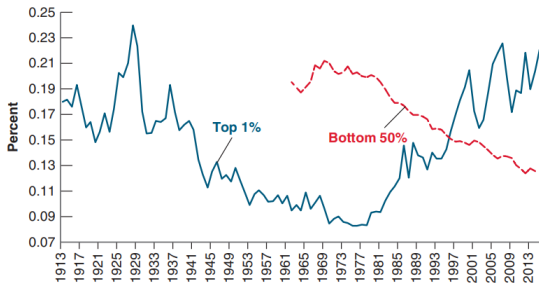
13.3 Growth, Churn, and Inequality

- Two common explanations for this phenomenon are increased international trade, which exposes low-skill U.S. workers to foreign competition, and **skill-biased technological progress**.
- The trade explanation, however, does not explain why the relative demand for high-skill workers seems to have increased even in those sectors not exposed to foreign competition.
- This observation, among other reasons, has led most economists to emphasize skill-biased technological progress as the primary explanation for increasing wage inequality.
- The process of growth that develops new goods and simultaneously makes other goods obsolete has been dubbed **creative destruction** by Joseph Schumpeter.

Inequality and the Top 1%

13.3 Growth, Churn, and Inequality

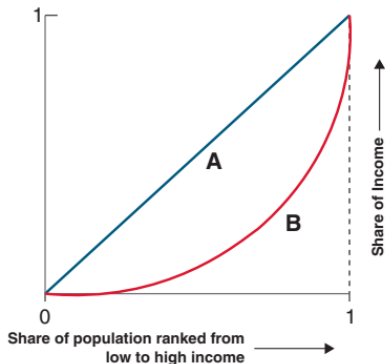
- Wage inequality is evident and appears to be driven by education and skill sets as one would expect.
- However, we are also seeing significant income inequality.
- The top 1% of income earners, relative to the bottom 50%, is shown in the following figure.



Gini Coefficient

13.3 Growth, Churn, and Inequality

- One way to measure the extent of the inequality is by computing a **Gini coefficient**.



The Gini coefficient is given by the ratio of area A to area $(A + B)$.

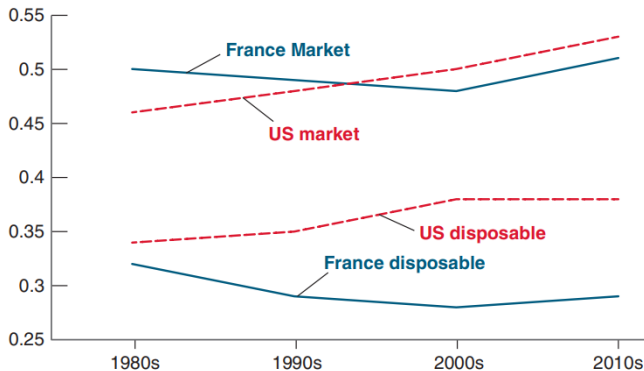
Growth and Inequality

13.3 Growth, Churn, and Inequality

- Given that inequality is increasing, the question for policy makers becomes "How do we achieve **inclusive growth**?"
- Current policies to reduce **disposable income inequality** include better education, minimum wage laws, and better governance rules within firms.
- In addition, countries with lower levels of disposable income inequality have higher levels of wealth redistribution.

The Evolution of Gini Coefficients in France and in the U.S. since the 1980s

13.3 Growth, Churn, and Inequality



13.3 Growth, Churn, and Inequality



Climate Change and Global Warming in Economics

13.4 Climate Change and Global Warming

- Markets do not function properly when externalities are present.
- One such major externality is the emission of greenhouse gases that have a societal cost but are not always considered when people make market choices.
- Given the potential damage due to increased global warming policy makers need to find some mechanism to combat this issue.
- Given the issue is a global issue, it needs to be addressed by all nations.
- However, it is extremely difficult to get an agreement across all countries.

Introduction of Carbon Tax

13.4 Climate Change and Global Warming

- One solution, proposed by William Nordhaus of Yale University, is to get countries willing to impose a carbon tax on goods imported from countries without a carbon tax.
- This policy would provide incentive for all nations to adopt a carbon tax to lower trade costs.